INTERNATIONAL MATHEMATICS CONFERENCE
Geometric Analysis by the Andes
September 7–11, 2026 ▪ Santiago, Chile

INVITED TALK / ABSTRACT

Stable Free-Boundary Minimal Hypersurfaces: Regularity, Compactness, and Existence

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Abstract

The Schoen–Simon theory gives fundamental regularity and compactness results for stable minimal hypersurfaces in closed Riemannian manifolds. In this talk, I will describe a free-boundary counterpart for hypersurfaces whose boundary lies on the ambient boundary and meets it orthogonally. Although the hypersurfaces satisfy this orthogonality condition, blow-ups at the ambient boundary exhibit two distinct flat models: the tangent hyperplane to the ambient boundary itself, corresponding to sheets collapsing onto the boundary, and a half-hyperplane meeting it orthogonally. I will explain intrinsic ε-regularity and sheeting theorems near both models. Their proofs combine geometrically adapted tilt functions, stability, and a De Giorgi-type iteration introduced by Bellettini. We also show that stationarity extends across singular sets of locally finite $\mathcal{H}^{n-2}$-measure. These results yield a compactness theorem for embedded stable free-boundary minimal hypersurfaces, with smooth graphical convergence away from a singular set of Hausdorff dimension at most $n-7$. As an application, we extend the free-boundary Almgren–Pitts existence theory to arbitrary dimensions, producing a free-boundary stationary integral varifold whose regular part is smoothly embedded and whose singular set has the optimal dimension bound. This is joint work with C. Bellettini, M. Li, and L. Sarnataro.

Keywords: stable free-boundary minimal hypersurfaces; regularity; compactness; Riemannian manifolds; ambient boundary; singular sets; Hausdorff dimension; free-boundary Almgren–Pitts existence theory